§9 Limitations, open problems, outlook for Phases 2 and 3 — part 1 of 3 (introduction; theoretical limits L1–L18)

Source: `paper/section9_limitations.tex`, lines 1–466 (`sec:limitations_and_outlook`: introduction `subsec:s9_intro`; `subsec:limitations_list`: L1–L18), at commit e6e12cd. Status: draft. §9 is presented in three parts: part 1 (1–466: introduction and L1–L18), part 2 (`09_limitations_2.md`, 467–846: delegations to Phase 1.α and open problems), part 3 (`09_limitations_3.md`, 847–1071: outlook for Phases 2 and 3, conclusion). §8 is `08_organizational_acs.md`. §9 defines no assumption. The Japanese manuscript is authoritative; this is a summary, not a translation, and adds no definition, assumption or result.

Introduction

`Reference: subsec:s9_intro`

The section discloses systematically the theoretical limits of the ACS theory constructed in Phase 1 (deterministic formulation; this paper), organizes the open problems, describes the research directions for Phases 2 and 3, and states the conclusion. It is not a section that adds a new constitutive requirement of an ACS, a new principle, TIPA mechanism, relation category, blockage or collapse condition, or applicability condition for the organizational ACS. It does not change the scope of the definitions, propositions and assumptions of earlier sections; it states what they do not claim under the present framework. The structure is: theoretical limits (`subsec:limitations_list`), delegations to Phase 1.α (`subsec:phase1a`), open problems (`subsec:open_problems`), outlook for Phases 2 and 3 (`subsec:phase_outlook`), and the conclusion (`subsec:conclusion`).

Theoretical limits

`Reference: subsec:limitations_list`

The Phase 1 boundaries stated explicitly from §2 to §8, and the parts that remain unsettled, are organized as L1–L18. Each item states its origin and the direction considered in Phase 2 or later.

L1: restriction to deterministic, discrete-time, compact spaces (`lim:L1`). Phase 1 is a framework centered on deterministic discrete-time state transitions and compact state, action and internal-model spaces. It therefore does not treat stochastic transitions, probability distributions, expectations, continuous-time limits, jump processes, unbounded spaces or variable-dimension state spaces as general theory. The restriction gives technical clarity in treating existence, closedness and compactness, but does not establish direct application to real systems that are stochastic, continuous-time or unbounded. *Origin:* §2 and the Phase 1 conventions of the whole paper. *Phase 2:* stochastic transitions, continuous time, local compactification and other weakenings.

L2: limited treatment of external-environment transition and reachability (environment dynamics) (`lim:L2`). §2 gives the environmental factors indispensable for persistence as the required environment space E and puts, as a background type, the external environment state space $E_{\mathrm{ext}} = E^{\mathrm{bg}} \times O$. Phase 1 only gives the transition of the action-response component, which can respond to the ACS's actions, as a map g_O on the side of the environment realization (`def:env_realization`). The generating law of the sequence of exogenous background components, reachability among environment states, stochastic variation of the environment, coupling dynamics with other ACS individuals, and co-evolution of the ACS and the environment are not formulated as independent environment dynamics. The possibility that different environment states exist is therefore distinguished from the dynamics by which such a state is actually reached. Reorganization of the self–environment boundary and time dependence of the external environment state space are treated in L16. *Origin:* the structural interpretation of the required environment space and the decomposition of the external environment state space (§2). *Phase 2:* generating laws of background components, reachability of environment transitions, stochastic environments, coupling dynamics with other individuals.

L3: cognitive processing capability is not formalized (`lim:L3`). Phase 1 does not define cognitive processing capability formally and does not introduce it as a single quantity. `rem:cognitive_capacity_policy` of §2 only states, as a description policy, that quantitative capacity, time responsiveness and quality of integration should be distinguished, that capacity and realized quantity should be distinguished, and that the bias that appears in how a capability is used, and its degree (bias strength), are not included in capability. Hence an index for comparing high and low capability, a measurement of the three divisions, and their relation to feasibility, blockage and collapse cannot be discussed from the results of Phase 1. *Origin:* §2, `rem:cognitive_capacity_policy`. *Phase 1.α / Phase 2:* formalization of the three divisions (for time responsiveness, the number of steps needed to reach a given criterion is a candidate) and their measurement.

L4: non-branching in individual identity (`lim:L4`). `ass:no_branching` allows at most one ACS-individual realization at time $t+1$ that is associated as the continuation of the same individual from a realization at time t, and excludes from the Phase 1 object the branching in which a copy or several successor realizations arise. For systems that include full replication, branching self-identity or inheritance of identity by several successor realizations, the relation of temporal self-identity cannot be applied as it stands. For an organizational ACS this corresponds to the case in which the successor realizations associated as continuation of the same organization become several, as in the split or spin-off of an organization. This non-branching is an assumption on individual identity, separate from the single-valuedness of the state transition map f. *Origin:* §2, `ass:no_branching`, `def:time_identity`. *Phase 2:* reformulation of temporal self-identity that permits branching, with temporal continuation relations among individual realizations, set-valued correspondences or genealogical structures as candidates.

L5: threshold calibration and inferential reach of T-STAB (`lim:L5`). T-STAB of §3 (`def:tstab`) is a three-principle time-integrated evaluation condition that integrates, on a finite time window, a lower bound of the SM evaluation, in-window satisfaction of local FR and EU, and one-step fluctuation of the SM evaluation. It does not mean Lyapunov stability, asymptotic stability or structural stability of dynamical systems theory. The lower reference value $\underline{s}_{\mathrm{SM}}$, the fluctuation bound $\delta_{\max}$ and the window width W_T (`def:tstab_params`) are reference parameters relative to the evaluation structure of the target ACS and are taken as given in Phase 1; no general quantitative calibration rule is given. This paper does not place a violation of T-STAB as a sufficient condition for blockage $A_{i,t} = \emptyset$, collapse $\mathrm{Col}_{i,t} = 1$, environment coupling with negative relation quality, complete isolation, or firing of Priority; connecting to them needs additional conditions. T-STAB is therefore a condition that describes and diagnoses, on a finite window, theoretical states related to the three principles, and does not by itself give general convergence, asymptotic stability or long-term prediction. Phase 1 also gives no independent infinite-time judgment condition for the long-term aspect of FR: `def:FR` gives local FR at each time and T-STAB evaluates its satisfaction within a finite window. This paper does not state, from T-STAB holding, the persistence of local FR outside the window or the maintenance of response diversity over infinite time as a general consequence. A full theory of the long-term aspect is listed in `subsec:phase1a`. *Origin:* §3, `def:FR`, `rem:sd_local_scope`, `def:tstab_params`, `def:tstab`, `prop:tstab_implies_principles`, `rem:conv_tstab_scope`, `rem:tstab_descriptive_status`. *Phase 2:* calibration rules for thresholds and window widths, parameter sensitivity, robustness, and implementation-specific conditional connections from T-STAB violation to the failure states.

L6: concrete implementation maps of TIPA and quantification of estimation error (`lim:L6`). §4 gives the functional structure and minimal requirements of M1, M2, M3, PSL and ISL, but does not uniquely fix the concrete functional forms of signal conversion, evaluation, Priority generation and response selection in each implementation. Phase 1 requires no universal error bound, convergence or agreement for the error between M1's state and environment estimates $\hat{x}_t$, $\hat{e}_t$ and the realized values x_t, e_t, or between M2's proxy evaluation and the principle-level truth (`rem:m1_estimate_nontruth`, `rem:error_bounds`). How small such errors stay in each implementation, their change over time, and their effect on the feasible action set or on reaching blockage are therefore not determined by Phase 1. TIPA gives a constructive, functional form in Phase 1 but is not a theory that uniquely identifies a particular biological mechanism or AI implementation. Also not fixed are: a mechanism that checks inconsistency between the structural separation B_{SE} and the self–environment classification of M1-b (`rem:xi_se_interpretation`); the uniqueness of the internal-model structure M that can correspond to the same f (`rem:M_independence`); the encoding map from the space of internal

representations $\mathcal{R}_{\text{int}}$ to $\mathcal{M}$ (rem:M_encodes_representation); and the concrete implementation forms of f_{auto}, f_{dir} and $\oplus$ (ass:f_decomposition). *Origin:* §4, rem:m1_estimate_nontruth, rem:error_bounds. *Phase 2:* concretization of implementation-specific maps, conditions for introducing error bounds, dynamics of estimation error, robustness and empirical calibration; the consistency-checking mechanism, the uniqueness of M and the implementation form of the decomposition of f are also tasks for Phase 2 or later. Introduction of the encoding map in a later Phase is not promised.

L7: abstractness of the update maps of §5 and undetermined coupling among variables (lim:L7). §5 treats $\boldsymbol{\beta}_t$, $d_{\text{eff},t}$, $\pi_{\tau,t}$, $\hat{\tau}_t$ and the four health quantities as values determined, by fixed readout rules, from the state, internal model and external environment state, or from a finite interval of operation records; their change from time t to $t+1$ is determined through f, Update and the realization of the external environment (the sequence of background components and g_O). No independent update map is placed. On the other hand, concrete learning rates, recovery rates, convergence speeds, interdependence functions among variables, long-term stability and general closed-form solutions are not fixed. $\hat{\tau}_t = \pi_{\tau,t}\tau$ is a derived quantity and has no independent update law. §5 therefore gives a minimal structure in the time direction, not detailed learning, degradation and recovery models of a particular implementation. *Origin:* §5. *Phase 2:* concrete update laws (including rules for updating the internal model from the difference between prediction and realization), coupling structure, and convergence, stability and recovery conditions, by model. For biases: concretizing the reference standard and scale of each bias, the readout method of bias strength, the description of formation by TIPA, PSL and ISL, the correspondence of combinations of formation to attribution divisions, identification of attribution divisions based on an observer model, the correspondence of implementation-level transformation and integration of biases to indices of the abstract structure description, and the expression of the direction and content of each bias (§5, def:bias_attribution, rem:beta_multiple_biases).

L8: non-causality of relation descriptors and implementation dependence of measurement and classification (lim:L8). The relation descriptors of §6 are an analytic form of classification C that describes the relation structure among objects and the changes of feasibility-related quantities observed in a relation-continuation window. The finite-window differences $\Delta_L \mathrm{sz}_i(A_{i,t})$, $\Delta_L d_{\text{eff},i,t}$, $\Delta_L \hat{\tau}_{i,t}$ do not mean a unique causal contribution of the object Y_j. The concrete construction of the local size indicator sz_i, category-judgment thresholds, durations, composite categories and causal attribution are not fixed as general theory, and a measure-theoretic refinement of the local size indicator (rem:finite_window_diff_limit) is not made. *Origin:* §6. *Phase 1.α:* detailing of classification and measurement rules. *Phase 2:* causal extensions including interventions and counterfactuals as needed, and the measure-theoretic refinement of the local size indicator.

L9: observability of blockage and collapse, recovery judgment and path dynamics (lim:L9). §7 keeps blockage $A_{i,t} = \emptyset$ apart from collapse $\mathrm{Col}_{i,t} = 1$ and defines the blockage path set, the collapse-reaching path set and the reaching times. It does not claim, however, that collapse can be judged in general from finite observation alone, and abstracts the implementation-dependent deterministic judgment rules of external recoverability. Inflow rates to path sets, reaching probabilities, mean reaching times and recovery probabilities are not defined in Phase 1. Also not given are an operational procedure that judges the value of the well-formed realization indicator $\mathrm{WF}_{i,t}$ from finite observation (def:wf_realization), and a general form of the judgment rules of (FT-1), (FT-2), (SC-1) and (SC-2) (def:functional_termination, def:collapse). *Origin:* §7. *Phase 2:* observability, judgment algorithms, analysis of stochastic reaching and recovery, quantification of recovery margin. The operational judgment procedure of $\mathrm{WF}_{i,t}$ is also a task for Phase 2. The judgment rules of (FT-1), (FT-2), (SC-1) and (SC-2) are treated in Phase 1 as fixed per target system.

L10: time-indexed record of the participating-ACS set and organizational identity dynamics are not formalized (lim:L10). §8 records the participating-ACS set $\mathbb{I}_{\text{org},t}(\mathcal{O})$ and number $n_{\text{org},t}(\mathcal{O})$ of an organization $\mathcal{O}$ at each time. A fixed finite candidate set P is an additional description only when the adopted model needs it and is not a general requirement of Phase 1; Phase 1 is therefore not a theory that fixes the number of members in time. On the other hand, a general transition law of the participating-ACS set itself, the dynamics of joining, leaving, generation and disappearance, and conditions for retaining organizational identity through those changes are not formulated. This paper does not state, from a change

in the participating-ACS number alone, holding or failing of qualification as an organizational ACS, collapse of the organization, or return of the same organizational ACS as a general consequence. *Origin:* §8, `def:org_participation`, `rem:org_qualification_loss` and `subsec:org_scope`. *Phase 2:* transitions of the participating set, joining and leaving, and the dynamics of organizational identity and return conditions, per target model.

L11: existence, construction and single-valued transition of the organization-level 7-tuple are not a general theorem (`lim:L11`). The organization-level ACS structure Σ_{org} of §8 is a structural form used when the ACS definition, constitutive requirements and well-formedness of §2 are satisfied for the organization itself. It is not proved that X_{org}, U_{org}, E_{org}, f_{org}, Π_{org}, M_{org}, $\text{Update}_{\text{org}}$ can be constructed as well-formed for an arbitrary social organization. The organizational state is not determined in general as a sum, average or product of individual states, and where an aggregation map is used it must be checked separately that a single-valued next organizational state follows from the same organizational state, action and external environment state. The applicability convention for the organizational ACS does not claim general existence or that any real organization qualifies. *Origin:* §8, `def:org_structural_form`, `rem:org_state_closure`, `def:org_acs`. *Phase 2:* existence conditions, well-formedness, single-valued transition, identifiability and empirical judgment for concrete organization models.

L12: judgment rules at the organization level and individual–organization aggregation (`lim:L12`). §8 applies the definitions of the three principles, blockage and collapse, time-integrated evaluation and path sets of earlier sections to the organization level only when a well-formed ACS structure holds for the organization itself (`rem:org_principles_conditional`). The application needs the judgment rules of (SC-1), (SC-2), (FT-1), (FT-2) to be fixed for the organization; their general form is not given, as in the individual case. A universal aggregation rule that converts individual feasibility, truth of the time-indexed satisfaction predicates of the three principles, relation descriptions, health quantities and so on into organizational quantities by mean, minimum, weighted mean or another operator is not given. Organizational-ACS qualification, feasibility, blockage, loss of well-formed realization and collapse are not identified with each other. The case in which successor realizations become several, as in the split of an organization, cannot be treated under the non-branching assumption (L4). *Origin:* §8, `rem:org_individual_nonimplication`, `rem:org_principles_conditional`, `rem:org_failure_scope`. *Phase 2:* individual–organization aggregation, organization-level three-principle evaluation, time integration, blockage and collapse paths and organizational dynamics as separate models.

L13: cross-domain correspondence and empirical validity are not verified (`lim:L13`). The paper intends an abstract structure applicable to humans and organisms, AI and robots, organizations and so on, but does not verify within the paper the measurability and empirical validity of the seven components, the three principles, TIPA, PSL and ISL, relation descriptors, and blockage and collapse conditions in individual domains. Formal applicability must therefore be distinguished from empirical validity. As for relations to existing theories, what the paper claims is consistency within the stated scope, not inclusion, superiority or generalization. Empirical methods to measure or estimate the quantity expressing the quality of updating of TIPA components (`def:update_quality`) are also not given. *Origin:* §1 and the scope of all sections. *Phase 2 and later:* domain-specific operationalization, measurement, comparative studies and examination of falsifiability.

L14: the full ECIT model and normative theory lie outside Phase 1 (`lim:L14`). The paper describes ethics not as good-and-evil judgment itself but as a structural constraint on feasibility preservation. Elements of ECIT such as love, wisdom and responsibility, the full conditions for firing ethical coordinative responses, the generation and change of social norms, concrete value choices and policy optimization are not claimed as formal consequences of Phase 1. These must be treated with a logical status separate from the descriptive theory of ACS. *Origin:* the restriction of §1 and §8 to descriptive theory and scope of application. *Phase 3:* the formal connection to ECIT, examined separately from the results of Phases 1 and 2.

L15: distinguishing success or failure of executing an action (`lim:L15`). Under the reading convention for the state transition map in §2 (`rem:f_reading`), the description of f does not distinguish an attempt of an action, a failure to execute it, and the degree of success of execution. Since no structure expressing executability of an action input separately from U is placed, the non-executability of an element of U at some time is absorbed into the next state given by f and is not represented as a formal object separate from that action not belonging to

the ACS's action space. The magnitude of the action difference (`def:action_difference`), multi-step action, comparison of contributions of actions and of the action-response component, and the balance of resource quantities common to the exterior and interior are also not quantified. *Origin:* §2, `rem:f_reading`. *Phase 2:* variables for success or degree of execution, and a structure that represents the actions executable at each time separately from U. Quantification of the magnitude of action difference and so on is reserved as a delegation to later Phases or as a candidate for future extension or reconsideration (the enumeration at the end of §2).

L16: time change of an individual's structure (`lim:L16`). Phase 1 treats the 7-tuple Σ_i of an individual as a time-independent structure; only the state, internal-model state and other time-indexed values change with time (§2, the convention on levels of description in `subsec:acs_7tuple`, `def:s_struct`). Expansion or reduction of the action space U itself (acquisition of new actions), change in the mechanism composition of the internal model, and change of the state space are therefore not described as time evolution of the same individual. Time dependence of the threshold that determines the feasible set (`def:Pi`) and the influence of the label structure of the action space U on its topology (`rem:action_space_structure`) are also not treated. Also not treated, although not components of the 7-tuple, are dynamic reorganization of the self–environment boundary (`rem:TSC_character`), a time-dependent external environment state space (`ass:Eext_time_fixed`), and a structure that maps individual-specific external environment state spaces to each other (`rem:Eext_relative`). Coupling dynamics with other ACS individuals are included in L2. *Origin:* §2, `subsec:acs_7tuple`, `rem:TSC_character`, `rem:Eext_relative`, `rem:action_space_structure`, `ass:Eext_time_fixed`, `def:Pi`. *Phase 2:* time-dependent individual structure and temporal self-identity under its change; extension to dynamic thresholds is a task for Phase 2 or later.

L17: operational identification, decomposition and judgment of required environmental factors (`lim:L17`). §2 gives a required environmental factor as a satisfaction-judgment map $\eta \in E$ and fixes its mathematical status by the constitutive requirement CR-1 (`subsec:environment_space`). It gives no procedure to identify a factor from an actual object and judge its satisfaction. Specifically not fixed are: the empirical identification procedure of required environmental factors and the treatment of identification error (`rem:necessity_identification`); the range and degree to which a factor whose external dependence appears only in some internal states is admitted as a factor; a procedure judging satisfaction of CR-1 from finite observation; the degree to which identification of a factor depends on the choice of external background (`rem:cr1_relativity`); additional conditions giving uniqueness of the decomposition of factors (`rem:factor_nonuniqueness`); and the granularity of classifying factors (`rem:eta_coarse_classes`). Existence of necessity is distinguished from identifiability. *Origin:* §2, `subsec:environment_space`, and the remarks cited. *Phase 2:* empirical identification procedures, identification error, the range and degree of external dependence, judging CR-1 by finite observation, dependence on the external background, and conditions for uniqueness of decomposition. Fixing the classification granularity is reserved as a delegation to later Phases or as a candidate for future extension or reconsideration.

L18: refinement of auxiliary quantities and analysis conditions related to the three principles (`lim:L18`). §3 defines auxiliary quantities only as far as needed for the time-indexed satisfaction predicates of the three principles and their conditional analysis. The following refinements are not made: an operational and causal decision criterion for identifying the environment-response component I_E, taking observation noise, time delay and internal feedback into account (`rem:env_response_observation`); directional analysis of local sensitivity in the x_E direction at boundary points (`rem:pi_E_local_sensitivity`); dynamic use of diagnostic indices that complement Π (`rem:diagnostic_role`); generalization of `lem:finite_exit` to history-dependent policies and conditions for uniform boundedness of exit time (`rem:uniform_bound`); the quantitative relation between the value of response diversity and the time to reach blockage (`rem:sd_degree`); the measure-theoretic refinement of μ_{eff} (`rem:mu_measure`); and sufficient conditions for deriving `ass:theta_exterior_blockage` from internal dynamics. *Origin:* §3 and the remarks and assumption cited. *Phase 2:* examination of each item above.

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